

Nicholas Lee

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Canadian Citizen

[Website](#) | [LinkedIn](#) | [GitHub](#)

Skills

Design & Manufacturing: SolidWorks (CSWA), AutoCAD, GD&T, DFM/DFA, PDM, FDM Printing, Sheet Metal, Mill, Lathe

Simulation & Analysis: COMSOL Multiphysics, ANSYS Discovery, CFD, FEA

Programming & Tools: Python, Excel, MATLAB/Simulink, SAP, C/C++, Ladder Logic (Siemens TIA Portal), Git

Work Experience

Research Intern

May 2026 – Aug. 2026

Harvard University, Aizenberg Lab

Cambridge, MA

- Increased simulated odorant exposure by 31% and peak signal by 42% by identifying optimal channel geometries and airflow patterns across a 247-case parametric grid of a nasal-inspired electronic nose model in COMSOL
- Determined optimal sensor placement per chemical species by measuring peak concentration position across a 100-point centerline probe, sweeping 10 chemical species of different molecular weights across 19 geometries
- Automated post-processing for 700+ transient simulation cases and 200K+ output rows using Python, Pandas, NumPy, and Matplotlib, generating 80+ figures and 16 tables
- Engineered watertight, 3D-printable airway models from CT-scanned animal skulls for future benchtop experiments

Undergraduate Research Assistant

Jan. 2026 – Apr. 2026

Multi-Scale Additive Manufacturing Lab

Waterloo, ON

- Delivered servo motor-driven two-stage gearbox demonstrators used in undergraduate lectures, iterating through FDM-printed gears, shafts, and housings across multiple print revisions until assemblies ran smoothly
- Accelerated helical gear design from an hour to 2 minutes per geometry change by building a parametric SolidWorks model driven by module, tooth count, and face width

Manufacturing Engineering Intern

Jan. 2025 – Aug. 2025

Refresco Beverages

Mississauga, ON

- Eliminated monthly piston failures on a can-rejection system by redesigning the pneumatic piston mount and cover in SolidWorks to shield the rod from syrup splash, delivering a full GD&T fabrication package for a third-party piston
- Integrated a new label-maker platform into plant layout drawings in AutoCAD for a \$1M line reformat, aligning maintenance, engineering, and production teams on the platform placement
- Initiated and secured plant-manager approval for a \$100K CAPEX project to reduce worker lifting injuries, scoping vacuum lift-assist solutions, evaluating vendor proposals, and presenting the recommendation to the management team
- Spearheaded standardization of the plant's preventive maintenance (PM) program, co-authoring 80+ PM procedures with task lists and BOMs in SAP and updating 100+ drawings, improving equipment reliability and maintenance efficiency

Student Design Teams

Mechanical Team Member – Interiors & Chassis

Sep. 2025 – Present

Midnight Sun Solar Car Team

Waterloo, ON

- Reduced ballast box mass by 20% with an aluminum sheet-metal assembly installed on the final competition vehicle, sized from hand calculations, designed in SolidWorks, and optimized as a flat pattern for waterjet cutting
- Designed surface-modeled cutting stencils that follow the aerobody's curvature, referenced to vehicle datums to standardize wheel-cover alignment and reduce cutting errors
- Simplified rear camera installation with a tool-free, press-fit 3D-printed mount, fit-tested across 5 print iterations

Mechanical Designer

Jan. 2026 – Present

Waterloo Automation Collective

Waterloo, ON

- Led the clamping subsystem of a student-built injection molder from concept through prototyping, building variable-driven parametric SolidWorks models to accelerate design iteration and milling the links for first assembly

Education

University of Waterloo

Expected May 2028

Bachelor of Applied Science in Mechatronics Engineering, Cumulative Average: 87%

Waterloo, ON

Relevant Coursework: Design & Dynamics of Machines, Electromechanical Machine Design, Sensors & Instrumentation, Mechanics of Deformable Solids, Fluid Mechanics

Leadership: Engineering Orientation Leader F26